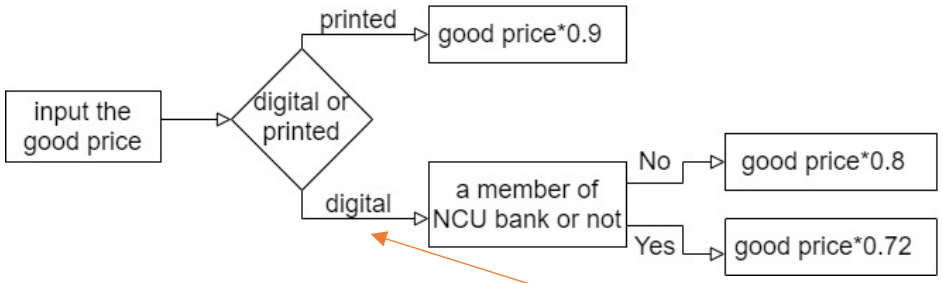
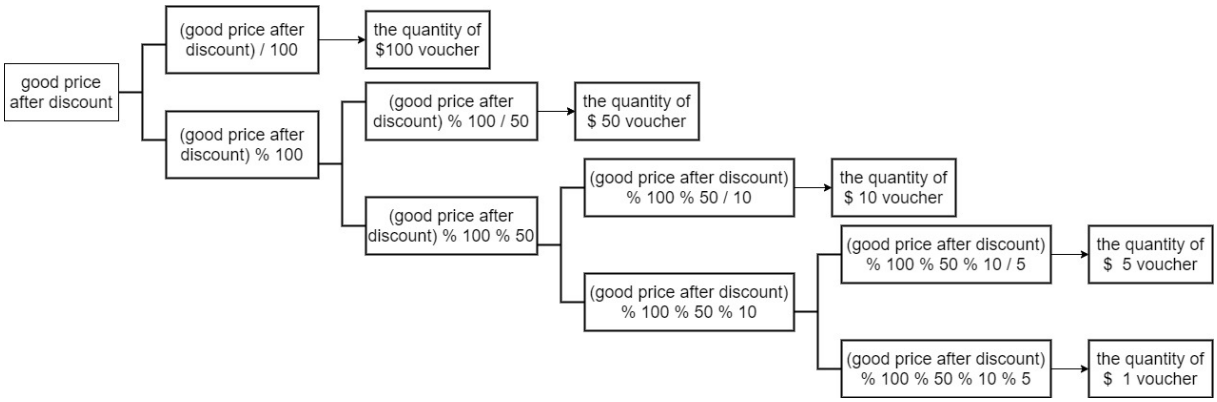
PA1: Quintuple Stimulus Vouchers Review

silver-kite-2a91

1. This is how my program works: You input the good price first, and then you will be given some options to decide how many discounts you get. Last, the voucher payment system will convert your final payment into accurate amounts of denominations.
   - Set your initial good price, and then the good price gets some discounts depending on what type of vouchers you select. Next, after you choose the digital one, you get another discount whether you are a member of the NCU bank.
   - My beginning thoughts were to calculate the quantities of the vouchers via keeping dividing the good price into different denominations. Until I ran into some special cases given by the teacher, my program gave some unwanted results and I found myself stuck. Later on, I went back to my code and detect some logic problems, and I will tell you how I solved the problem with a “rob Peter to pay Paul ” tactic in the following, which is a strategy by keeping adding functions to the code without changing the whole project due to limited time. After all, I’m glad but not able to spend all day long writing code.

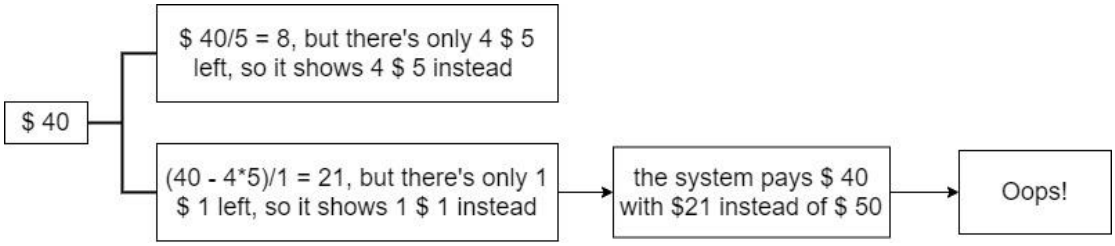


Maybe you may question why did I not multiply the price by 0.9 at the digital part, that’s because the system would round down the good price to an integer so that in sample 1, you will not be paying $ 568 but $ 567.



This is how my program works. You enter the good price and the system asks what type of vouchers do you prefer, and then it asks whether you are a member of NCU bank if you stored it as a digital type. Later, you will get your final good price after several discounts. Next, the system calculates the quantities of each voucher to satisfy the payment by constantly dividing the remainder of the good price after discount by every denomination of voucher from the largest to the smallest, then you will get the numbers of each voucher you should pay with.

- I had to admit that this strategy was so intuitive that I believe that almost everyone had come up with this kind of solution at first. However, here comes the interesting part. In sample 1, this solution does not work on the last part, when you are going to pay $ 40 with 1 $ 50, 4 $ 5, and 1 $ 1. Since $ 40 is less than 50, the system shows that 40/50 = 0, so rather than paying it with $ 50, it takes the money left, which are 4 $ 5 and 1 $ 1, resulting in paying $ 40 with $ 21, which is not what we want. (By the way, I'm lazy to change my written code back to the flawed one and demonstrate the wrong result, so please allow me to show you the logic with flow diagram)



2. After several modifications, all samples announced on the bulletin board worked through my code.
3. The hardness I encountered was the special requirement in the payment. I got stuck for a while for my program does not function properly when my remaining vouchers can't pay the exact price, which I believe was indeed a challenge to most students in class. For instance, while my total amount of money can pay the price, however, it gave me an output less than the actual price I should pay (please refer to the previously mentioned). After digesting the problem, I came up with a new solution to solve the problem. (please read 3.5) I knew that my new method would face some difficulties in the approaching future and conflict under some weird payments, however, I am ready for it and will find a better strategy to deal with them.

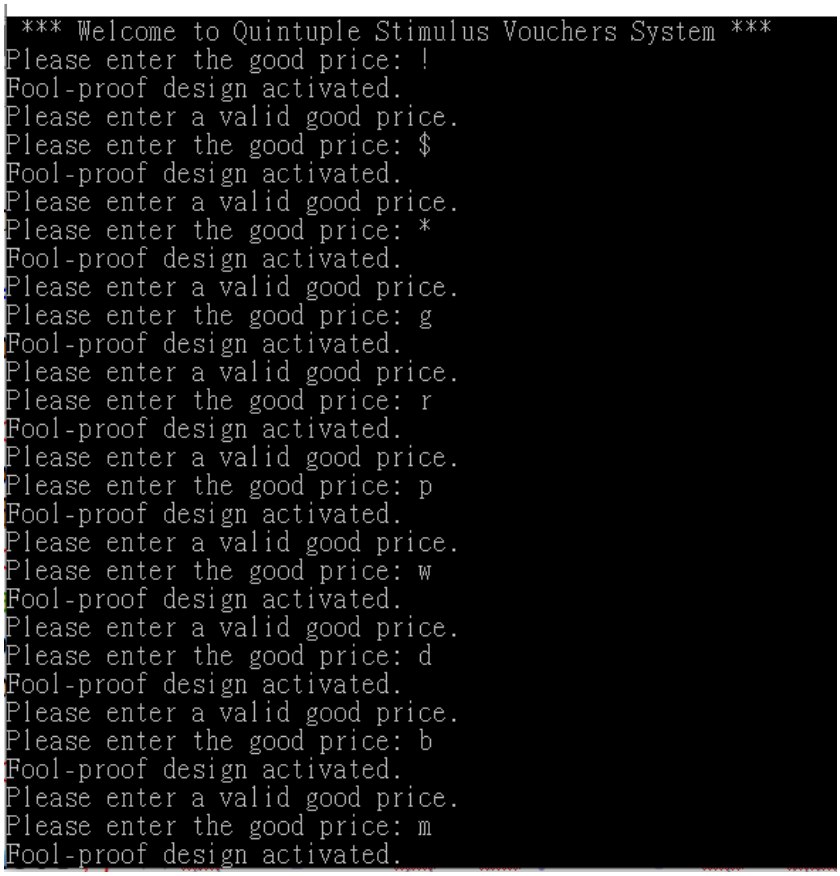
3.5 How I fix my code

This is what I do to my code. Take sample 1 as an example, when paying $ 40 with 1 $ 50, 4 $ 5, and 1 $ 1, I wrote if the sum of the smaller denomination is

less than the price to be paid, the system will take the larger one before the smaller ones and set the remaining good price to 0 so that the remaining vouchers will not be count in payment mistakenly.

4. I did not create an extra version of the assignment. In my opinion, the extra version consists of a whole different perspective, therefore, what I'm doing can't be counted as a bonus version; instead, what I'm doing is optimizing. Since the negative price was invalid to the payment, in the good price section, if the system reads in -1, you will get kicked out by the system; meanwhile, if you enter a number less than 0, I use the function "goto", and you will be asked to enter the good price again instead of kicking you out of the system to keep the code running. Another interesting function is that if you input something instead of a number, you will also be asked to enter the good price again.

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More bonus but meaningful functions still designing…

5. To be honest, I was having a hard time getting used to this kind of teaching. Facing the problem without any tools, in the beginning, my mind went blank and I have no idea what I can do. After I learned some programming during class, it took me a long time to get acquainted with those pieces of stuff, so it had been a while until I started to work on my assignment. Fortunately, spending much more time on the assignment and consulting some classmates, I

had successfully compiled and executed my program. In my opinion, what sets PBL (project-based learning) apart from conventional teaching is that PBL enforces you to start from 0 to 1, which inspires the student to learn by the teacher but more by themselves, and sparks their learning potence. I hope that in the next classes, the professor gives more clearer instructions on the application of the functions taught in class so that I will know how to use what I learned in my coming assignments more precisely.

6. Update: after I uploaded my assignment, I opened my code again with codeblock and found out that my comments(註解) have all been garbled(亂碼), however, open my code with dev c++ displays normally. Hence, I put my code down here directly to show you the comments.

```
#include <iostream>
#include <string>
using namespace std;

int main()
{
	int goodprice, hundred_dollars, fifty_dollars, ten_dollars, five_dollars, one_dollars, a, b;
	hundred_dollars = 5 ;
	fifty_dollars = 5 ;
	ten_dollars = 5 ;
	five_dollars = 5 ;
	one_dollars = 5 ;
	a = 0 ;
	b = 0 ;
	int remain;
	remain = 830;//餘額
	string vouchers, member, str1 = "d", str2 = "p", str3 = "y", str4 = "n";

	cout << " *** Welcome to Quintuple Stimulus Vouchers System *** " << endl;
	while(remain > 0)
{
	label:cout << "Please enter the good price: " ;
	cin >> goodprice;
	if (goodprice == -1)
		{cout << "Bye~";return 0;}
```

```
            else if (goodprice < 0)
            {goto label;} //除了-1 以外輸入任何小於 0 的數字都會請你重新輸
入
            else{}
        cout << "Please enter the type of vouchers (d/p): " ;
        cin >> vouchers;

        if (vouchers.compare (str1) == 0)
            {goodprice = goodprice * 1 ;
            cout << "Are you a member of NCU Bank (y/n): " ;
            cin >> member;
            if (member.compare (str3) == 0){ goodprice = goodprice * 0.72 ;}
                else if (member.compare(str4) == 0){ goodprice = goodprice *
0.8;}}
                else if (vouchers.compare(str2) == 0){goodprice = goodprice *
0.9;}
                else {goodprice = goodprice * 1;} //比較字串是否相等來決定折
扣
        remain = remain - goodprice ; //餘額會拿來做為一些判斷的依據,所以要
時時紀錄餘額
        if (remain < 0)
            {cout << "The price you need to pay: " << goodprice << endl;
            cout << "You don't have enough voucher to buy the good." << endl <<
endl << endl;
            remain = remain + goodprice ;
            goto label;} //付不出錢時,讓系統把款項加回去餘額
            else{}
        if (one_dollars*1 + five_dollars*5 + ten_dollars*10 + fifty_dollars*50 +
hundred_dollars*100 >= goodprice)
        {
            cout << "The price you need to pay: " << goodprice << endl;
            cout << "The payment is processed as follows" << endl;
        if (goodprice == 0);
            else if(0 < hundred_dollars && hundred_dollars <= goodprice/100)
            {goodprice = goodprice -    hundred_dollars*100;
            cout << "        " << hundred_dollars << " $100 voucher" << endl;
            hundred_dollars = 0 ;} //超過 100 元的數量時則全部拿出來付
            else if (goodprice < 100 && goodprice > fifty_dollars*50 +
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```
ten_dollars*10 + five_dollars*5 + one_dollars*1 && goodprice <
hundred_dollars*100)
            {
                if (hundred_dollars > 0)
                    {while (goodprice/100 < hundred_dollars)
                    {a = a + 1 ;
                    goodprice = goodprice/100 ;
                    hundred_dollars = hundred_dollars - 1 ;
                    cout << "        " << a << " $100 voucher" << endl;}}
                    else if (fifty_dollars > 0)
                    {while (goodprice/50 < fifty_dollars)
                    {b = b + 1 ;
                    goodprice = goodprice/50 ;
                    fifty_dollars = fifty_dollars - 1 ;
                    cout << "        " << b << " $ 50 voucher " << endl;}}
            }
            else if (hundred_dollars > goodprice/100)
            {cout << "        " << goodprice/100 << " $100 voucher" << endl;
            hundred_dollars = hundred_dollars - goodprice/100 ;
            goodprice = goodprice%100 ;} //張數要大於 0 才能拿來支付
      if (goodprice == 0);
            else if (0 < fifty_dollars && fifty_dollars <= goodprice/50)
            {goodprice = goodprice - fifty_dollars*50 ;
            cout << "        " << fifty_dollars << "$ 50 vouchers" << endl;
            fifty_dollars = 0 ;} //超過 50 元的數量時則全部拿出來付
            else if (goodprice < 50 && goodprice > ten_dollars*10 + five_dollars*5
+ one_dollars*1 && goodprice < fifty_dollars*50 + hundred_dollars*100)
            {
                if (hundred_dollars > 0)
                    {while (goodprice/100 < hundred_dollars)
                    {a = a + 1 ;
                    goodprice = goodprice/100 ;
                    hundred_dollars = hundred_dollars - 1 ;
                    cout << "        " << a << " $100 voucher" << endl;}}
                else if (fifty_dollars > 0)
                    {while (goodprice/50 < fifty_dollars)
                    {b = b + 1 ;
                    goodprice = goodprice/50 ;
```

```
                fifty_dollars = fifty_dollars - 1 ;
                cout << "        " << b << " $ 50 voucher " << endl;}}
        }
        else if (fifty_dollars > goodprice/50)
        {cout << "        " << goodprice/50 << " $ 50 voucher" << endl;
        fifty_dollars = fifty_dollars - goodprice/50 ;
        goodprice = goodprice%50 ;} //張數要大於 0 才能拿來支付

    if (goodprice == 0);
        else if (0 < ten_dollars && ten_dollars <= goodprice/10)
        {goodprice = goodprice - ten_dollars*10 ;
        cout << "        " << ten_dollars << " $ 10 voucher" << endl;
        ten_dollars = 0 ;} //超過 10 元的數量時則全部拿出來付
        else if (goodprice < 10 && goodprice > five_dollars*5 + one_dollars*1
&& goodprice < ten_dollars*10 + fifty_dollars*50 + hundred_dollars*100)
        {
            if (hundred_dollars > 0)
                {while (goodprice/100 < hundred_dollars)
                {a = a + 1 ;
                goodprice = goodprice/100 ;
                hundred_dollars = hundred_dollars - 1 ;
                cout << "        " << a << " $100 voucher" << endl;}}
                else if (fifty_dollars > 0)
                {while (goodprice/50 < fifty_dollars)
                {b = b + 1 ;
                goodprice = goodprice/50 ;
                fifty_dollars = fifty_dollars - 1 ;
                cout << "        " << b << " $ 50 voucher " << endl;}}
        }
        else if (ten_dollars > goodprice/10)
        {cout << "        " << goodprice/10 << " $ 10 voucher" << endl;
        ten_dollars = ten_dollars - goodprice/10 ;
        goodprice = goodprice%10 ;} //張數要大於 0 才能拿來支付
    if (goodprice == 0);
        else if (0 < five_dollars && five_dollars <= goodprice/5)
        {goodprice = goodprice - five_dollars*5 ;
        cout << "        " << five_dollars << " $    5 voucher" << endl;
        five_dollars = 0 ;} //超過 5 元的數量時則全部拿出來付
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```
        else if (goodprice < 5 && goodprice > one_dollars*1 && goodprice <
five_dollars*5 + ten_dollars*10 + fifty_dollars*50 + hundred_dollars*100)
        {
            if (hundred_dollars > 0)
                {while (goodprice/100 < hundred_dollars)
                {a = a + 1 ;
                goodprice = goodprice/100 ;
                hundred_dollars = hundred_dollars - 1 ;
                cout << "        " << a << " $100 voucher" << endl;}}
                else if (fifty_dollars > 0)
                {while (goodprice/50 < fifty_dollars)
                {b = b + 1 ;
                goodprice = goodprice/50 ;
                fifty_dollars = fifty_dollars - 1 ;
                cout << "        " << b << " $ 50 voucher " << endl;}}
        }
        else if (five_dollars > goodprice/5)
        {cout << "        " << goodprice/5 << " $    5 voucher" << endl;
        five_dollars = five_dollars - goodprice/5;
        goodprice = goodprice%5 ;} //張數要大於 0 才能拿來支付
    if (goodprice == 0);
        else if (0 < one_dollars && one_dollars <= goodprice)
        {goodprice = goodprice - one_dollars ;
        cout << "        " << one_dollars << " $    1 voucher" << endl;
        one_dollars = 0 ;} //超過 1 元的數量時則全部拿出來付
        else if (one_dollars > goodprice)
        {cout << "        " << goodprice << " $    1 voucher" << endl;
        one_dollars = one_dollars - goodprice ;} //張數要大於 0 才能拿來支
付

    if (remain > 0){cout << "The remaining vouchers are as follows: " << endl;}
        if (hundred_dollars > 0)
        {cout << "        " << hundred_dollars << " $100 voucher" << endl;}
        else{}
        if (fifty_dollars > 0)
        {cout << "        " << fifty_dollars << " $ 50 voucher" << endl;}
        else{}
        if (ten_dollars > 0)
```

```
            {cout << "        " << ten_dollars << " $ 10 voucher" << endl;}
            else{}
            if (five_dollars > 0)
            {cout << "        " << five_dollars << " $    5 voucher" << endl;}
            else{}
            if (one_dollars > 0)
            {cout << "        " << one_dollars << " $    1 voucher" << endl;}
            else{}
            cout << endl << endl; //一大堆空白只是為了符合格式
        }
}
        if (remain == 0)
            {cout << "You do not remain any voucher. Bye~" ; return 0;} //餘額其實是跟消費券同步額度以及數量,所以這裡是拿餘額來判斷五券是否花完
        return 0;

}
```